
MACRO REGIMES, FUNDAMENTALS & THE REACTION FUNCTION

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v1.0 · September 20, 2026

Abstract

Macro context explains why markets move; it rarely tells you *when*. This paper builds the bridge between the two honestly. We construct a growth×inflation regime classifier from public data with one pre-committed rule — the sign of the three-month change in year-over-year industrial production and CPI — and measure asset behavior in each of the four quadrants with confidence intervals, cell counts, and an explicit non-stationarity flag; over 1948–2026 the four regimes split almost evenly (24–27% each) and last a median of two months, which is itself a warning against regime-timing. We decode the Taylor rule term by term, reproduce it on core PCE and an Okun-law output gap, and read the *gaps* between rule and policy as the story: the fixed-coefficient rule prescribes a funds rate 2.68 percentage points above the actual as of mid-2026, and it says the opposite about the Volcker 1980s — a deliberate counterexample. We decompose the Chicago Fed's financial-conditions index and show, with overlapping-window caveats, that it is a strong *state* descriptor but a weak *return* timer. We make the fundamentals bridge quantitative — earnings, margins, the equity risk premium, and the Fed-model debate presented from both sides — with a worked rates-to-multiple-compression example, and we ship a one-page reproducible macro dashboard. Every number uses free data and is reproducible from the accompanying code.

Keywords — Macro regimes, growth and inflation quadrant, reflation, goldilocks, stagflation, deflation, Taylor rule, reaction function, dual mandate, financial conditions index, NFCI, equity risk premium, Fed model, multiple compression, real-time data, data vintages, look-ahead bias, non-stationarity

1 Introduction

1.1 Motivation and thesis

Every trader carries a macro story. Rates are going up, so growth stocks are done; inflation is sticky, so the Fed can't cut; conditions are loose, so buy the dip. These stories are not worthless — they organize a chaotic world into a handful of states, and the states are real. But the same stories are responsible for more confident, expensive mistakes than almost anything else a discretionary trader does, because they smuggle a *timing* claim inside a *context* claim. "We are in a stagflation regime" is a description. "Therefore sell equities this month" is a forecast, and the forecast is far weaker than the description sounds.

The thesis of this paper is a single sentence, and the rest of the document is its defense: *macro is a superb lens for understanding the distribution of outcomes and a poor lever for timing them, and the honest way to use it is to build regime labels and reaction-function estimates you can actually compute from public data, attach confidence intervals and counterexamples to every claim, and treat the residual — the gap between the rule and the world — as the only part worth trading on.* We will build each of those objects: a growth×inflation regime classifier, a Taylor-rule estimate of where policy "should" be, a financial-conditions aggregate, a fundamentals bridge from rates to multiples, and a one-page dashboard that ties them together. And at every step we will do the thing that macro commentary almost never does — report the number even when it embarrasses the story, and name the period in which the story failed.

1.2 Scope, prerequisites, and intended reader

This paper sits in the Trading Library's *Macro & Sentiment* track (Context pillar). It assumes the macro-foundations paper: the four master variables (growth, inflation, policy, liquidity), the business cycle, the mechanics of a central bank, the key data releases, and cross-asset transmission. It assumes the *Statistics for Traders* material carried over from the Quant track: means, standard errors, confidence intervals, base rates, and the difference between a coincident description and a predictive claim. It assumes you can run a short Python script that pulls a public series and computes a return; it assumes no econometrics beyond a correlation and a t-style interval.

After reading this paper, a reader can:

- classify any month since 1948 into a growth×inflation regime from two free FRED series, state the one arbitrary rule that does it, and defend every arbitrary choice inside that rule;

- read a table of asset returns by regime *with* confidence intervals, cell counts, and a non-stationarity flag, and explain why the wide, overlapping intervals are the finding rather than a nuisance;
- write the Taylor rule from memory, decode every term, compute it on public data, and interpret the rule-minus-policy gap as a regime signal rather than a law;
- decompose a financial-conditions index, explain why it is a cleaner single summary of the macro-to-markets channel than any one spread, and show with data why it still does not time monthly returns;
- value the equity index from earnings, margins, and a discount rate; state the Fed-model argument and the standard critique of it fairly; and put a number on how much a 100 bp move in rates compresses a fair multiple;
- build and read a one-page macro dashboard from FRED and Yahoo; and
- recite, without prompting, the reasons a macro-timing backtest is probably lying to them — vintages, revisions, overlapping windows, and a sample of only a handful of genuinely independent cycles.

1.3 Data and reproducibility

All numerical examples and figures use freely available data — FRED for macro series (industrial production, CPI, core PCE, unemployment, the natural rate, fed funds, Treasury yields, the National Financial Conditions Index, and the NBER recession indicator) and the Yahoo chart API for prices (^GSPC, TLT, GLD). Each figure caches its inputs under `figures/data/` so it reproduces offline, and each is generated by a named script under `figures/`. Where a figure uses a single classification rule, that rule is fixed *in advance* and never re-tuned to improve the picture; reporting a rule chosen after seeing the result would be exactly the data-snooping this paper warns against. *This is a research and educational document, not investment advice.*

One discipline binds every causal statement in this paper: every directional causal claim is shipped with a counterexample or an explicit period in which it did not hold. Regimes are wonderful for describing the shape of outcomes and treacherous for predicting them, and the counterexamples are how we keep ourselves honest about the difference.

1.4 Organization of this paper

The paper is complete: all sections below are written, and this subsection is a map of the finished arc rather than a promissory note.

- **Section 2 Regime classification you can actually compute** builds the growth×inflation quadrant from two public series, states every arbitrary choice, and measures asset behavior

per quadrant with intervals, cell counts, and a non-stationarity flag (Figures 1 and 2).

- **Section 3 The central-bank reaction function** decodes the Taylor rule term by term, reproduces it on public data, explains what "data-dependent" operationally means, and reads the rule-minus-policy gap as the signal (Figure 3).
- **Section 4 Financial conditions as the aggregate** decomposes the NFCI, argues why an aggregate beats any single variable as a state descriptor, and shows honestly that it is a weak monthly return timer (Figure 4).
- **Section 5 The fundamentals bridge** values the index from earnings, margins, and the equity risk premium, presents the Fed-model debate from both sides, and makes rates→multiple compression quantitative with a worked example.
- **Section 6 Building a macro dashboard** ships a one-page reproducible FRED/Yahoo dashboard across cycle, inflation, rates, conditions, and equities, with the current read (Figure 5).
- **Section 7 Macro → volatility and flows** maps regime onto the volatility factors of the volatility-surface paper and the flow calendar of the flow-landscape paper, and treats event-volatility around scheduled releases.
- **Section 8 Limits and humility** confronts the lagging-and-revised nature of regime labels, the vintage problem (ALFRED versus FRED), overlapping windows, the tiny effective sample of independent cycles, and the specific reasons macro-timing backtests lie.
- **Section 9 Common misconceptions, glossary, references, and changelog** close the paper.

2 Regime classification you can actually compute

The word *regime* does a lot of quiet work in market commentary, and most of it is unearned. A regime is only useful if two people looking at the same data would assign the same label — which means the label must be a *computation*, not a *vibe*. This section builds the most common macro regime scheme, the growth×inflation quadrant, as an explicit rule you can run on free data, and then measures what actually happened to assets inside each quadrant, with the honesty machinery (intervals, counts, non-stationarity) that the description usually lacks.

2.1 Two axes, four quadrants

The scheme has two axes because two forces dominate the discount rate and the cash flows behind almost every asset: the *direction of growth* and the *direction of inflation*. Cross them and you get four cells, each with a folk name that a macro trader will recognize:

	Inflation accelerating	Inflation decelerating
Growth accelerating	Reflation — expansion with rising prices	Goldilocks — expansion with cooling prices
Growth decelerating	Stagflation — slowing with rising prices	Deflation — slowing with cooling prices (a disinflationary slowdown)

The intuition behind the names is standard cross-asset transmission (the macro-foundations paper): reflation tends to favor real assets and cyclicals; goldilocks — growth *and* falling inflation — is the friendliest state for long-duration equities and bonds together; stagflation is the state that punishes the classic 60/40 because stocks and bonds fall at once; and the disinflationary-slowdown quadrant favors duration (long bonds) as the market prices cuts. These are the *stories*. Section 2.4 tests them against data, and the test is deliberately unkind.

2.2 The one rule, and every arbitrary choice inside it

To make the label a computation we must commit to proxies and a threshold, *in advance*, and then never move them to make the picture nicer. Here is the entire rule, with each arbitrary choice named as arbitrary:

- **Growth proxy:** industrial production (FRED INDPRO). *Arbitrary:* industrial production is a narrow, goods-tilted slice of a services-dominated economy; real GDP is broader but quarterly and heavily revised, and payrolls or the PMI would give different turning points. INDPRO is chosen because it is monthly, long, and rarely benchmark-revised beyond recognition — not because it is the "true" growth.
- **Inflation proxy:** headline CPI (FRED CPIAUCSL). *Arbitrary:* core CPI, core PCE (the Fed's actual target), or trimmed-mean measures would each shade the label differently, especially around energy spikes.
- **From level to direction:** take the year-over-year rate of each series, then take the *three-month change* of that year-over-year rate. Positive means the annual rate is *accelerating*; negative means *decelerating*. *Arbitrary:* the three-month window is a choice; one or six months would relabel a meaningful minority of borderline months. Using the *momentum* of the YoY rate (rather than its level) is a deliberate choice to capture *turning points* rather than *high-versus-low* — a distinction that matters, because "inflation is high" and "inflation is rising" are different states.
- **The cut:** the label is the sign pair. Growth-momentum > 0 and inflation-momentum > 0 is Reflation; > 0 and ≤ 0 is Goldilocks; ≤ 0 and > 0 is Stagflation; ≤ 0 and ≤ 0 is Deflation. Exact ties fall to the "decelerating" branch and, with floating-point YoY values, essentially never bind.

Two honesty notes belong here, before any result. First, the classifier uses *final-vintage* data — today's fully revised INDPRO and CPIAUCSL — so the historical labels are what we *now* know each month was, not what a trader could have known *then*. Section 8 is devoted to why that gap matters and how much of the apparent signal it can manufacture. Second, because the rule splits on the *sign* of two roughly symmetric momentum series, it is almost mechanically going to produce four cells of similar size; a near-even split is therefore not a discovery about the economy but a property of the construction, and we flag it as such rather than dress it up.

```
# The entire classifier, in words made code (see figures/fig_e2_quadrant.py):
# yoy[m]      = X[m]/X[m-12] - 1          # year-over-year rate
# mom3[m]     = yoy[m] - yoy[m-3]        # 3-month change of the YoY rate
# g, i       = mom3(INDPRO), mom3(CPI)   # growth- and inflation-momentum
# label      = Reflation if g>0 and i>0
#              Goldilocks if g>0 and i<=0
#              Stagflation if g<=0 and i>0
#              Deflation if g<=0 and i<=0
```

2.3 What the label looks like through history

Run the rule from the first month it can be computed (1948-04, once both series have twelve months of history plus a three-month lookback) to the latest common month (2026-07): 936 monthly labels. Figure 1 shows them as a color barcode above the two momentum series that generate them.

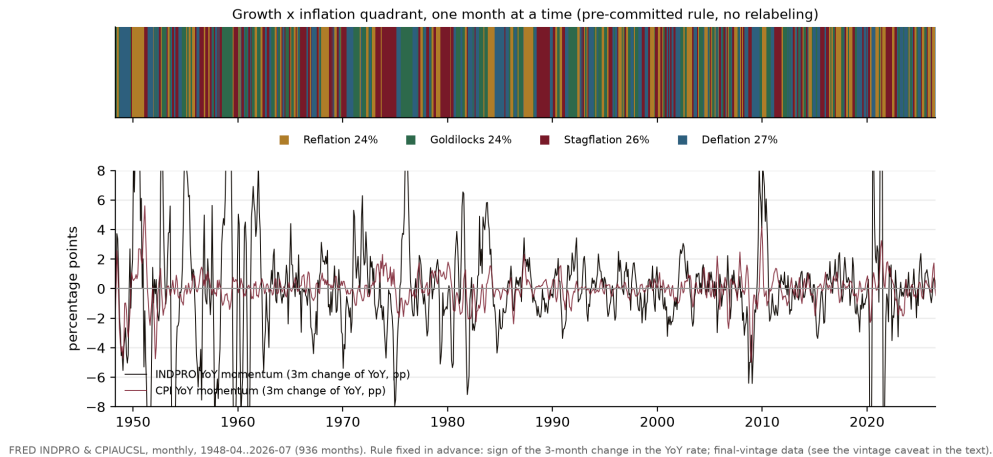


Figure 1. Growth×inflation regime, one month at a time, 1948-04 to 2026-07 (936 months). The top strip colors each month by quadrant under the fixed sign-of-three-month-momentum rule; the panel below plots the two momentum series (growth in dark, inflation in red) whose signs produce the labels. The four regimes occupy 23.5% (Reflation), 24.0% (Goldilocks), 25.7% (Stagflation) and 26.7% (Deflation) of months — a near-even split that is a property of the construction, not a finding.

Reproduce with figures/fig_e2_quadrant.py.

The share of months per quadrant is **Reflation 23.5% (n = 220)**, **Goldilocks 24.0% (n = 225)**, **Stagflation 25.7% (n = 241)**, and **Deflation 26.7% (n = 250)** (`fig_e2_quadrant.py`). As warned, they are almost equal — the sign-split guarantees roughly this. The genuinely useful statistics are about *persistence*, and they carry a hard lesson for anyone who wants to trade regimes. The probability that next month carries the same label as this month is **61.6%**, and the mean spell length is **2.60 months** (median 2, maximum 17). In words: the regime you are in will most likely still be there next month, but it turns over on a two-month timescale, and long, clean regime runs are the exception. A classifier that flips every two months on average is a poor foundation for a slow-moving allocation, and a superb way to generate whipsaw if you trade each flip. The current month (2026-07) is **Deflation** — growth-momentum -0.30 pp and inflation-momentum -0.48 pp, i.e. both the growth and the inflation rate were decelerating on this measure — which the dashboard in Section 6 will corroborate with the raw series.

2.4 Asset behavior per quadrant — with the honesty attached

Now the part everyone wants: how did assets *do* in each regime? We match each quadrant label to the same month's total-price return for three free proxies — the S&P 500 (^GSPC, from 1927), long Treasuries (TLT, from 2002), and gold (GLD, from 2004) — and report the mean, a 95% confidence interval, and the cell count. Figure 2 shows the equity-and-cross-asset picture; the full table, including the tradable-lag variant, is printed below.

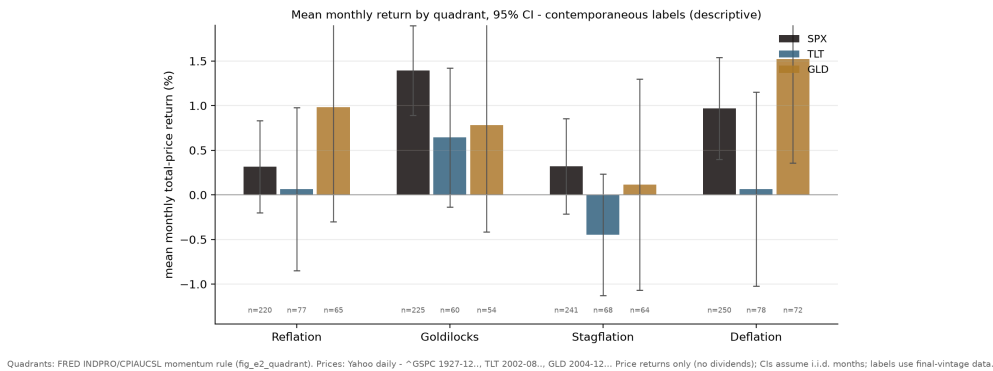


Figure 2. Mean monthly price return by growth×inflation quadrant with 95% confidence intervals, contemporaneous labels. S&P 500 (dark, 1948-2026), long Treasuries (blue, TLT 2002-2026), gold (gold, GLD 2004-2026). Cell counts are printed under each bar; the confidence intervals are wide and overlapping across quadrants, which is the finding. Reproduce with `figures/fig_e2_quadrant_assets.py`.

The full table of mean monthly return (%), 95% CI, and n per cell (`fig_e2_quadrant_assets.py`):

Asset	Reflation	Goldilocks	Stagflation	Deflation
SPX	+0.32 ± 0.52 (n=220)	+1.39 ± 0.50 (n=225)	+0.32 ± 0.53 (n=241)	+0.97 ± 0.57 (n=250)
TLT	+0.06 ± 0.91 (n=77)	+0.64 ± 0.78 (n=60)	−0.45 ± 0.68 (n=68)	+0.06 ± 1.09 (n=78)
GLD	+0.98 ± 1.28 (n=65)	+0.78 ± 1.20 (n=54)	+0.12 ± 1.18 (n=64)	+1.52 ± 1.17 (n=72)

Read the table the way the story predicts, and much of it *rhymes*: equities do best in Goldilocks (+1.39%/month, the highest cell), gold does best in the disinflationary slowdown (+1.52%/month), and long bonds do worst in Stagflation (−0.45%/month, the only negative equity-or-bond cell of note). None of these signs would surprise a macro textbook. But now read the *intervals*, which is where the discipline lives. The equity Goldilocks edge over Stagflation is +1.39% versus +0.32% — about +1.07 pp/month — and each cell's 95% interval is roughly ±0.5 pp, so the two overlap. Overlapping one-sample intervals do *not*, however, imply the difference is insignificant — that is a common fallacy; the proper check is a direct two-sample test of the difference, not whether either interval contains the other's mean. Tested that way the equity gap is significant at conventional levels (a two-sample test gives $t \approx 2.88$, $p \approx 0.004$), so it is better described as established-but-modest than as merely suggestive. The bond and gold cells are worse still: intervals of ±0.7 to ±1.3 pp around means of similar magnitude mean that *most cross-quadrant differences for TLT and GLD are statistically indistinguishable from zero*. The barchart's honest headline is not "regimes drive returns"; it is "regimes shift the *odds* a little, inside a great deal of noise."

Two flags are mandatory here. **Cell-count flag**: the task's threshold for a fragile cell is $n < 15$; *no cell in this table falls below 15* (the smallest is GLD-Goldilocks at $n = 54$), so no cell is individually starved — but that is the *good* news and it is not the binding constraint. **Non-stationarity flag**: the binding constraint is that TLT and GLD exist only from 2002 and 2004, so every bond and gold cell is drawn entirely from the post-2001 disinflationary, quantitative-easing era — *one* macro epoch. A "Deflation" cell for gold is not a statement about deflationary slowdowns in general; it is a statement about the specific 2004–2026 disinflations, which were accompanied by falling real rates and a secular gold bull market. The SPX cells span 1948–2026 and cross several genuinely different monetary regimes, which is why they deserve marginally more trust — but even there, as Section 8 argues, the *effective* number of independent regime episodes is a handful, not 936.

2.5 The tradable version, and why the signal shrinks

There is a subtler trap hidden in the contemporaneous table: month t 's label uses month t 's INDPRO and CPI, which are not published until month $t+1$. Matching them to month t 's return is therefore *look-ahead* — descriptive of "what the regime looked like while it was on," but not something a

trader could have acted on. The implementable version matches month t 's return to the label of month $t-2$, the freshest label actually computable at the start of month t given the roughly one-month publication lag. Here is that lagged ($t-2$) table:

Asset	Reflation	Goldilocks	Stagflation	Deflation
SPX	$+0.66 \pm 0.56$	$+1.05 \pm 0.48$	$+0.28 \pm 0.59$	$+1.01 \pm 0.50$
TLT	$+0.33 \pm 0.66$	-0.20 ± 0.97	$+0.62 \pm 0.96$	-0.48 ± 0.98
GLD	$+1.07 \pm 1.09$	$+0.80 \pm 1.35$	$+0.82 \pm 1.37$	$+1.04 \pm 1.02$

Compare the two SPX rows. Contemporaneously, Goldilocks beat Stagflation by +1.07 pp/month; on the implementable lag the same gap is $+1.05 - 0.28 = +0.77$ pp, and Deflation (+1.01%) now nearly ties Goldilocks. For TLT the *signs flip* in three of four cells between the two tables — the clearest possible sign that the bond-by-quadrant pattern is noise being read as structure. This is the general rule and it deserves to be stated plainly: **making a macro signal implementable — by respecting the publication lag — reliably shrinks it, because part of what looked like predictive power was the look-ahead of using data before it existed.** And this lagged table *still* ignores revisions, so it *overstates* the truly-real-time signal (Section 8). The counterexample to "regimes predict asset returns," then, is built into the method: the more carefully we make the classifier tradable, the less it predicts.

2.6 Using the label without fooling yourself

The operational way to use a regime label is not as a trade trigger but as a *prior on the distribution*. A compact recipe: (1) compute the current label *and* its momentum trajectory — a Reflation reading whose growth-momentum is already rolling over is closer to Stagflation than a fresh one, so read the derivative, not just the level; (2) treat the label as shifting the *odds and the tails*, not the direction — a Stagflation reading is a reason to expect diversification to fail (stocks and bonds down together, the -0.45% /month TLT cell of Section 2.4) and to size defensively, not a reason to short outright; (3) never act on the contemporaneous label, because it is look-ahead — use the lagged, publication-respecting version (Section 2.5) for anything tradeable, and remember it still ignores revisions; (4) weight each cell by the width of its evidence — the SPX cells (1948–2026, several monetary epochs) deserve more trust than the TLT/GLD cells (one post-2001 epoch); and (5) treat every regime call as one of a *handful* of independent episodes, so a striking backtest across "all Stagflation months" is really one or two cycles in disguise (Section 8.3). Used this way the classifier is a lens on *which risks are elevated and which hedges are likely to work* — a genuinely valuable input — rather than a signal generator — a role Figure 2's mostly wide intervals caution against, even though the single equity Goldilocks-minus-Stagflation gap, tested

with a proper two-sample test rather than by eyeballing interval overlap, is itself significant ($t \approx 2.88$).

3 The central-bank reaction function

If regimes are the state of the world, the central bank is the largest single actor reacting to that state, and its reaction moves the discount rate under every asset. A *reaction function* is a rule that maps the state of the economy to the policy rate. No central bank literally obeys one, but a good reaction function does two things at once: it gives you a *nowcast* of what the bank is likely to do, and — more valuable — it gives you a *benchmark* against which the bank's actual choice can be read as unusually tight or unusually easy. This section builds the canonical reaction function, the Taylor rule, decodes it, reproduces it on public data, and argues that the *gap* between rule and policy is the tradeable object, not the rule itself.

3.1 The Taylor rule, decoded term by term

Taylor's 1993 rule, in its original form, is (Taylor 1993):

$$i = r^* + \pi + 0.5 (\pi - \pi^*) + 0.5 y$$

where i is the prescribed nominal fed funds rate, r^* is the neutral real rate, π is current inflation, π^* is the inflation target, and y is the output gap (percent by which real output exceeds its potential). Decoding each term in plain words:

- r^* (*Taylor used 2*): the *neutral real rate* — the inflation-adjusted rate that neither stimulates nor restrains an economy running at potential. It is the anchor: with inflation at target and output at potential, the rule prescribes $i = r^* + \pi^* = 2 + 2 = 4\%$ nominal. Everything else is a deviation from this resting point.
- $+\pi$: add current inflation, converting the real anchor into a nominal rate. This is the *Fisher* piece — to hold a given *real* stance, the nominal rate must move one-for-one with inflation. A rule that did *not* include this term would ease in real terms exactly when inflation rose, which is destabilizing.
- $+0.5 (\pi - \pi^*)$: the *inflation-gap* response. For every point inflation runs above its 2% target, add half a point *on top of* the Fisher pass-through. Combined with the $+\pi$ term, the nominal rate rises by **1.5** for every 1 point of inflation — the famous *Taylor principle*: to be stabilizing, nominal rates must rise *more* than one-for-one with inflation, so that the *real* rate rises when inflation does.
- $+0.5 y$: the *output-gap* response — the second half of the dual mandate. When output is 1% above potential (an overheating economy), add half a point; when it is below, subtract. This

is the term that lets the rule *cut* into a recession even when inflation has not yet fallen.

The two 0.5 coefficients are Taylor's original weights, and we keep them fixed — no tuning to fit any particular decade. The rule's beauty is that it encodes the *dual mandate* (price stability and full employment) as a single decision: the inflation-gap term is the price-stability leg, the output-gap term is the employment leg, and the two 0.5s are the bank's implied relative weight on each.

3.2 Reproducing it on public data

To compute the rule we need public proxies for its two gaps. For inflation we use *core PCE* year-over-year (FRED PCEPILFE), because core PCE strips the food-and-energy noise the Fed looks through when it reads the trend — a defensible choice for a policy rule, though the 2 percent target itself is written on *headline* PCE, and the swap introduces its own arbitrariness (headline versus core, PCE versus CPI). The output gap is harder, because "potential output" is unobservable. We use the standard shortcut: *Okun's law* relates the output gap to the unemployment gap with a coefficient of about 2, so we proxy $y \approx 2 \times (u^* - u)$, where u^* is the CBO natural rate of unemployment (FRED NROU, quarterly, forward-filled to months) and u is the unemployment rate (UNRATE). Substituting into the 0.5 y term, the unemployment gap enters the rule with coefficient 1.0. We compare the rule's prescription to the effective fed funds rate (FEDFUNDS, monthly average). Every input is final-vintage — the real-time caveat (Orphanides 1998, 2001) is central and returns in Section 8. Figure 3 shows the result from 1960 to 2026.

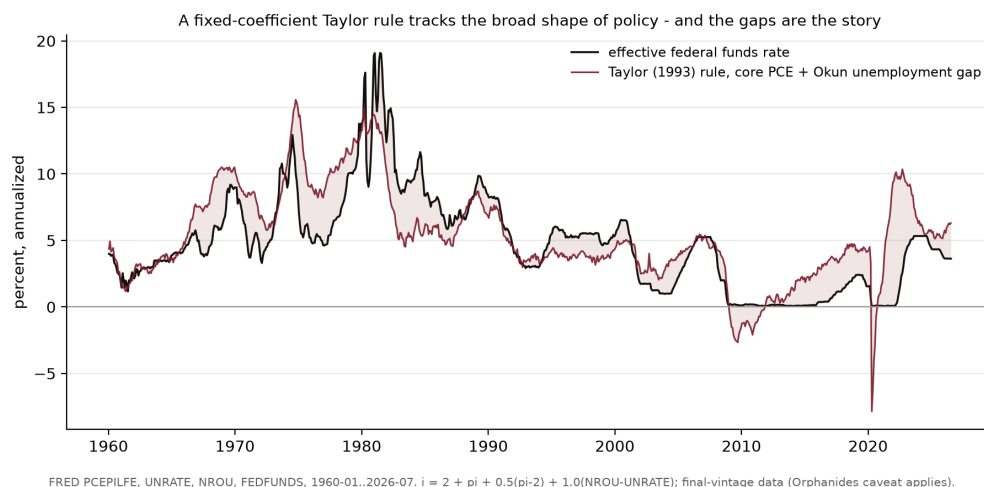


Figure 3. A fixed-coefficient Taylor rule versus the effective federal funds rate, 1960-2026. The rule (red) uses core PCE inflation and an Okun-law unemployment gap with Taylor's original 0.5/0.5 weights; the actual funds rate is dark. Shaded areas mark the rule-minus-policy gap. The rule tracks the broad shape of policy (correlation 0.749) but the gaps are large, persistent, and regime-dependent — they, not the level, are the story. Reproduce with `figures/fig_e2_taylor.py`.

The rule tracks the broad contour of six decades of policy: the correlation between rule and actual is **0.749**, and the mean absolute gap is **1.95 pp** (`fig_e2_taylor.py`). A single fixed equation, with coefficients set in 1993 and inputs anyone can download, explains three-quarters of the co-movement of the funds rate over sixty years — a genuinely impressive fact, and the reason the Taylor rule is the default reaction-function benchmark. But 1.95 pp of average absolute gap is not small; a benchmark that is off by two points on average is a benchmark whose *deviations* carry the information.

To make the computation concrete, build the rule term by term for the latest month (2026-07) from the public inputs. Core PCE inflation was $\pi = 3.34\%$ year-over-year; the CBO natural rate was 4.39% against an actual unemployment rate of 4.10%, an unemployment gap of +0.29 pp (output fractionally *above* potential by the Okun proxy). The rule assembles as: neutral real rate $r^* = 2.00$, **plus** inflation $\pi = +3.34$ (*the Fisher term*), **plus** the inflation-gap response $0.5 \times (3.34 - 2.00) = +0.67$, **plus** the output-gap term $1.0 \times (4.39 - 4.10) = +0.29$ — *summing to 6.31%*. The effective funds rate that month was 3.63%, so the gap is +2.68 pp, and the decomposition shows where it comes from: *almost all of it is the inflation terms — core PCE running 1.34 points above target contributes $+3.34 + 0.67 = +4.01$ of the 6.31%, while the near-closed labor gap adds only +0.29*. A trader reading this learns something the single gap number hides: the pressure on policy is an inflation story, not a labor* story, so the release that will actually move the rule — and therefore rate expectations and event-volatility — is the inflation print, not the jobs report.

3.3 The gap is the signal — read with a counterexample

The rule-minus-policy gap, decade by decade: 1960s +1.10 pp, **1970s +2.67 pp**, **1980s -2.28 pp**, 1990s -0.88 pp, 2000s +0.36 pp, 2010s +1.30 pp, **2020s +2.87 pp**. A positive gap means the rule prescribes a *higher* rate than the Fed set — i.e. policy was *easier* than the rule; a negative gap means policy was *tighter*. This decade profile is a compact monetary history. The 1970s ran a large *positive* gap: the rule wanted much higher rates than Burns delivered, and the Great Inflation followed — the textbook case that policy was persistently too easy. The single largest gap in the whole sample is **+9.93 pp in March 2022** (rule 10.13% versus an actual 0.20%): with inflation surging and unemployment low, the fixed rule screamed for double-digit rates while the Fed was still at the zero bound, about to begin the fastest hiking cycle in forty years. As of the latest common month (2026-07) the rule prescribes **6.31% against an actual 3.63%, a gap of +2.68 pp** — the fixed rule still reads policy as easier than a mechanical inflation-and-labor response would dictate, consistent with a Fed that judges much of the 2022–2023 inflation to have been transitory supply shock rather than a durable overheating. One structural caveat travels with any such gap: the rule holds the neutral real rate fixed at Taylor's original $r^* = 2.00$, and that anchor is doing more work than it looks — it alone contributes 2.00 of the 6.31 prescription. Contemporary time-varying estimates put r^* well below 2% for much of the recent sample; the

Laubach–Williams and Holston–Laubach–Williams models place it nearer 0.5–1% (Laubach & Williams 2003; Holston, Laubach & Williams 2017), and substituting such an estimate lowers the prescribed rate by more than a percentage point and shrinks the gap accordingly. The *level* of the gap is therefore a statement about the assumed neutral rate as much as about policy; the regime-dependent *shape* is the robust part.

Now the mandatory counterexample, because "the rule says policy is too easy, therefore the Fed will hike" is exactly the kind of causal shortcut this paper forbids without a caveat. The **1980s carry a mean gap of –2.28 pp** — policy was, by the fixed rule, *2.28 points too tight for a decade*. Volcker deliberately overtightened relative to any mechanical rule to rebuild the Fed's anti-inflation credibility, and it worked: inflation broke. A trader who in 1982 had shorted bonds because "the Taylor rule says rates should be lower, so the Fed will cut" would have been early and wrong for years, and would have missed that the *point* of the regime was a Fed choosing to sit far from the rule. The fixed-coefficient rule did *not* hold as a policy predictor across the 1980s; it held as a description of how far the Fed had chosen to depart from mechanical behavior. That is the correct way to use the gap: as a measure of *policy stance relative to a neutral benchmark*, whose sign tells you the *direction of the pressure* on the bank — not as a countdown timer to the next move.

One further honest wrinkle: during the 2009–2015 zero-lower-bound years the mean gap was only **+0.12 pp**, but the *rule itself went negative in 40% of those months*. The gap understates how easy the Fed wished to be, because the funds rate could not follow the rule below zero — which is why the era's easing happened through the balance sheet (quantitative easing) rather than the rate. A reaction-function benchmark built on the policy *rate* is blind to policy conducted through *quantities*, and Section 7 and the liquidity discussion in the macro-foundations paper are where that blindness is repaired.

3.4 "Data-dependent," nowcasting, and the market-implied gap

Central bankers describe themselves as *data-dependent*, which sounds like a refusal to commit to a rule but operationally means something precise: the coefficients of their implicit reaction function are *state-contingent*. In a credibility-building regime (Volcker) the inflation-gap weight is effectively much larger than 0.5; in a deflation-scare regime (2010s) the output-gap weight dominates and the inflation term is allowed to run soft. "Data-dependent" is therefore not the absence of a reaction function — it is a reaction function whose weights the bank reserves the right to change, and the trader's job is to nowcast *which weights are live now*. A practical nowcast of the next move stacks three things: (1) the fixed-rule prescription and its gap (this section); (2) the *trajectory* of the two gaps — a rule racing ahead of policy, as in 2021–2022, precedes hikes, while a rule falling below policy precedes cuts; and (3) the officials' own words, which reveal the current weights. Instantiated on today's readings: the fixed rule sits +2.68 pp above policy (leg 1,

a hawkish-leaning gap), but the *trajectory* matters more — the gap has been *narrowing* as the Fed normalizes toward the rule rather than widening as in 2021 (leg 2, which points away from urgent hikes), while officials' framing of much of the 2022–2023 inflation as supply-driven keeps the effective inflation-gap weight below the mechanical 0.5 (leg 3). The three legs disagree, and the disagreement is the point: it is precisely when the fixed rule, the gap trajectory, and the communicated weights *conflict* that the market's priced path is most fragile and a single data surprise carries the most repricing power.

The most tradeable version of all this is the gap between *market-implied* and *rule-implied* rates. The market prices a forward path of the funds rate (via fed funds futures and OIS); the rule implies its own path from projected inflation and unemployment; and central-bank communication implies a third. When these three disagree, the disagreement is where policy surprises — and therefore volatility — are priced. The rule-minus-policy gap in Figure 3 is the *backward-looking* cousin of this: it tells you how far current policy sits from a neutral benchmark, which sets the *direction* in which a data surprise is most likely to move expectations. A large positive gap (rule far above policy, as now) means the asymmetry is toward *hawkish* surprises; a large negative gap means the risk is a *dovish* repricing.

4 Financial conditions as the aggregate

Growth, inflation, and the policy rate are three separate dials. What ultimately reaches markets is their *joint* effect on the price and availability of money — credit spreads, equity valuations, funding rates, volatility, the dollar. A *financial-conditions index* (FCI) is an attempt to collapse all of that into one number. This section takes the cleanest public one apart, argues why an aggregate is a better single macro-to-markets summary than any component, and then shows with data that "better summary" does *not* mean "return-timing signal."

4.1 What an FCI is, and why aggregate beats any single spread

The Chicago Fed's National Financial Conditions Index (NFCI, FRED NFCI) is a weekly index built from **105** individual measures of risk, credit, and leverage — money-market spreads, bond and equity-market volatility, credit-default swaps, survey-based lending standards, and more — combined so that the index has mean zero and unit standard deviation over its history (Brave & Butters, Chicago Fed methodology). By construction, **zero is the historical average**, positive readings are *tighter than average*, and negative readings are *looser than average*. The methodology's key move is that the components are combined to extract the common factor they share, so idiosyncratic noise in any single spread is averaged out.

The Chicago Fed also publishes the NFCI's *decomposition* into three subindexes, which is where the word "decomposed" earns its keep: a **risk** subindex (volatility and funding-risk measures

across markets), a **credit** subindex (credit conditions and spreads), and a **leverage** subindex (debt and equity leverage, which tends to *lead* — leverage builds quietly in good times and is the kindling for the next stress). A companion series, the *adjusted* NFCI (ANFCI), strips out the part of financial conditions mechanically explained by the current state of the cycle and inflation, isolating the piece that is *unusual given* the macro backdrop — the cleaner object for the question "are conditions loose or tight *relative to where the economy is?*" The decomposition matters for reading a move: a tightening driven by the leverage subindex (deleveraging) is a different animal from one driven by the risk subindex (a volatility spike), and only the aggregate-plus-decomposition lets you tell them apart.

That averaging is why an aggregate is the cleaner macro-to-markets link than any one variable. Any single measure — the VIX, the high-yield spread, the 2-year yield — is a mix of *the common financial-conditions signal* and *its own idiosyncratic story* (a VIX spike from one event, a spread move from one sector). The index cancels the idiosyncrasies and keeps the common part. So as a *description of the current state of financial stress and easing*, the NFCI is more reliable than staring at any one gauge — it will not be fooled by a single spread that widened for a reason unrelated to the macro cycle. That is a real and useful property, and it is the honest form of the claim "financial conditions are the aggregate."

4.2 A weak return timer — the honest test

The seductive next step is: "loose conditions are good for stocks, tight conditions are bad, so trade the NFCI." We test exactly this, with a rule fixed before looking at returns — loose = $\text{NFCI} < 0$, tight = $\text{NFCI} \geq 0$ (the split point is the index's own historical mean, not a fitted threshold) — and a timing convention that respects publication (the NFCI for a week ending Friday is released the following Wednesday, so we enter the first S&P 500 session at least five calendar days after the week-end and hold 21 trading days). Figure 4 and the numbers below are the result over 1971–2026.

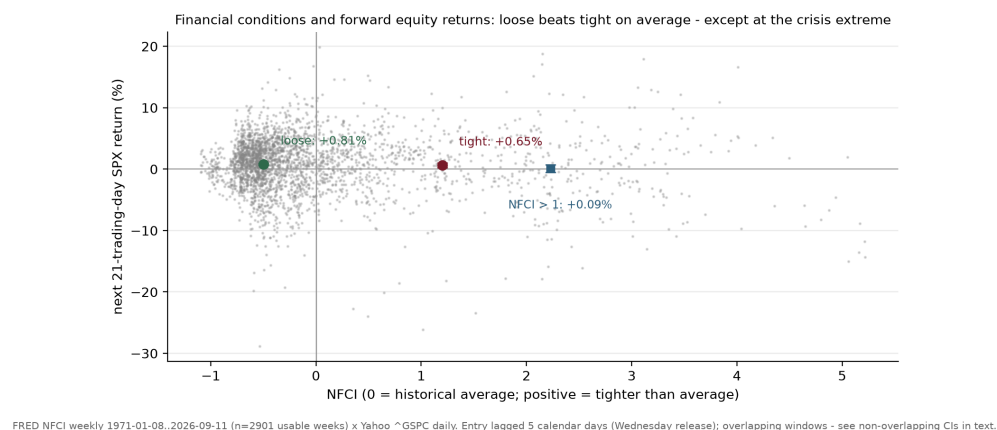


Figure 4. Chicago Fed NFCI versus the subsequent 21-trading-day S&P 500 return, 1971-2026 (2,901 usable weeks). Grey dots are individual weeks; the markers show the mean forward return in the loose (NFCI<0, green) and tight (NFCI≥0, red) states with 95% confidence intervals, and the extreme-tightness tail (NFCI>1, blue). Loose beats tight on average by only 0.16 pp per month, with heavily overlapping intervals; the one robust feature is the depressed forward return at the crisis extreme.

Reproduce with figures/fig_e2_nfcipy.

The means with confidence intervals for the loose/tight split (fig_e2_nfcipy): **loose (NFCI < 0): +0.806% ± 0.175 over 21 trading days (n = 2,050 weeks); tight (NFCI ≥ 0): +0.649% ± 0.368 (n = 851).** Loose does beat tight — but by only **+0.157 pp per month**, and the intervals overlap heavily. The correlation between the NFCI *level* and the forward 21-day return is **−0.055** — statistically distinguishable from zero on 2,901 observations, but economically a rounding error. And the overlapping-window problem inflates even that thin significance: adjacent weekly windows share most of their 21-day horizon, so the effective sample is far smaller than 2,901. When we recompute on a *non-overlapping* subsample (every fifth week), the ordering nearly reverses: loose +0.710% ± 0.402 (n = 410) versus tight **+0.749% ± 0.852** (n = 171). On independent windows, tight-conditions months were followed by *slightly higher* average returns than loose-conditions months — the opposite of the story, well inside the noise.

This is the counterexample, and it is decisive: **"tight financial conditions lead to lower equity returns" did not hold on independent monthly windows over 1971–2026.** The mechanism is not mysterious. The NFCI is *coincident* with stress, not leading it — conditions are tightest *at* the bottom, right before the strong recovery returns, so "buy when tight" often beats "buy when loose." The single feature that *does* survive is the crisis extreme: when the NFCI is above **1** (a genuine funding crisis — 2008, March 2020), the forward 21-day return averages only **+0.093% ± 0.631** (n = 378), materially below both the loose and the tight means. Extreme tightness is informative — not because it predicts weakness, but because it flags the states in which the return *distribution* has a fat left tail. The NFCI's job, then, is the one Section 4.1 gave it: a superb *state descriptor* and *risk gauge*, and a poor *monthly return timer*. Using it as the former is macro done well; using it as the latter is the mistake this section exists to prevent.

5 The fundamentals bridge

Regimes and reaction functions move the *inputs* to asset prices; fundamentals are how those inputs become a *level*. This section builds the bridge for the equity index — from earnings and margins to a discount rate and a multiple — presents the one genuinely contested piece (the "Fed model") from both sides, and makes the most practically important mechanism, rates→multiple compression, quantitative with a worked example.

5.1 What the index is worth: earnings, a payout, and a discount rate

The value of the equity index is the present value of the cash it will return to owners. The compact form is the *Gordon growth* model: for a stream of shareholder cash flows growing at rate g and discounted at rate r ,

$$P = D / (r - g), \text{ equivalently a justified forward multiple } P/E = b / (r - g),$$

where D is next period's shareholder cash flow (dividends plus net buybacks), E is earnings, $b = D/E$ is the *payout ratio* (the share of earnings returned), g is the long-run nominal growth rate of those cash flows, and r is the required nominal return on equity. Three fundamentals feed this: (1) *earnings*, the denominator's engine; (2) *margins*, because earnings = revenue $\times$ margin and revenue grows roughly with nominal GDP, so a durable change in the profit share of the economy changes the sustainable earnings base; and (3) the *discount rate* r , which is the risk-free rate plus the equity risk premium. Each deserves a sentence of realism. Earnings at the index level are cyclical and, at peaks, flatter themselves; *margins* are the most mean-reverting series in finance over long horizons — corporate profits as a share of output oscillate around a slowly-moving mean, and extrapolating a record margin is a classic valuation error (Damodaran, public valuation data). A concrete version of the trap: suppose net margins sit at a cyclical-high 13% against a long-run mean nearer 10%. An analyst who capitalizes today's earnings at a normal multiple is implicitly assuming a 13% margin *forever*; if margins mean-revert toward 10% over the coming years, the earnings base shrinks by roughly $(10/13 - 1) \approx -23\%$ *even with revenue flat*, and the "cheap" multiple on peak earnings was an illusion. This is exactly why cyclically-adjusted measures — averaging earnings over a full cycle, as in the Shiller CAPE — exist: they blunt the peak-margin, peak-earnings flattery. The most dangerous input in equity valuation is therefore often not the discount rate but the *sustainability of the margin* feeding earnings. And g cannot durably exceed nominal GDP growth for the whole index, however tempting it is to plug in a single company's growth.

5.2 The equity risk premium

The *equity risk premium* (ERP) is the extra return investors demand for holding stocks instead of the risk-free asset: $r = r_f + \text{ERP}$. It is the single most important — and least observable — number in equity valuation. Two ways to get at it: the *historical* ERP (the realized average excess return of stocks over bonds, which is backward-looking and sensitive to the sample) and the *implied* ERP (the discount rate that makes a forward cash-flow model equal today's index price, given consensus earnings and a risk-free rate) (Damodaran; Ilmanen 2011). The implied ERP is the more useful of the two because it is forward-looking and updates daily, but it inherits every assumption about future earnings and growth that go into the model — a high implied ERP can mean "stocks are cheap" or "the earnings forecasts are too high," and the model cannot tell you which. Ilmanen's *Expected Returns* is the standard synthesis of how the ERP, the bond risk premium, and other premia vary with the cycle and with valuations (Ilmanen 2011): premia are *higher* when valuations are low and conditions are fearful, and *lower* after long bull markets — the same "buy when it hurts" asymmetry the NFCI tail showed in Section 4. A back-of-envelope implied ERP anchors the idea. Take a forward earnings yield near 4.7% (a forward P/E around 21×, illustrative), subtract a real risk-free rate of about 2.0% (the 4.68% nominal 10-year of Section 6 less roughly 2.3–2.7% expected inflation), and, in the crudest single-stage form, add expected long-run real earnings growth of about 2%: the implied ERP lands near $4.7\% + 2.0\% - 2.0\% \approx 4.7\%$, comfortably inside the 3.5–6% band that has held historically. The number is not the point; the point is that "the ERP" is a *derived* quantity whose value swings with whichever earnings-yield, inflation, and growth inputs you feed it — so a responsible ERP claim reports its assumptions, not just its answer.

5.3 The Fed model, honestly, from both sides

The *Fed model* is the popular heuristic that the equity index is fairly valued when its forward *earnings yield* (E/P) equals the nominal 10-year Treasury yield — "stocks are cheap when the earnings yield exceeds the bond yield." It has an honest case and a decisive critique, and an honest treatment requires both.

The case for. As a *comparative-value* heuristic, it captures a real competition: bonds and stocks compete for the same capital, and when bonds yield more, the required return on stocks (and hence their earnings yield) tends to rise too. Empirically, over roughly 1965–2000 the forward earnings yield and the 10-year yield tracked each other closely, and the model "worked" as a rough fair-value gauge for a generation of practitioners. That co-movement was real; it is why the heuristic became conventional wisdom.

The critique (the standard one). Asness's *Fight the Fed Model* (Asness 2003) makes the decisive objection: the model commits a *money-illusion* error by comparing a **real** quantity to a

nominal one. The earnings yield is a real quantity — earnings grow roughly with inflation, so E/P already compensates for inflation — while the Treasury yield is nominal, embedding an inflation premium. Setting them equal implicitly demands that stocks yield *more* in real terms when inflation is high, which is backwards. Asness shows two things with data: (1) the historical correlation that made the model "work" was an artifact of the specific 1965–2000 period, when high inflation dragged *both* nominal bond yields and equity earnings yields up together — a *spurious* co-movement driven by inflation, not a valuation law; and (2) once you separate them, it is the *earnings yield alone* (a valuation measure) that predicts long-run real equity returns, while the bond-yield comparison adds *nothing* to that prediction. The counterexamples are explicit: in the 1930s–1950s stocks persistently yielded *more* than bonds for reasons of risk, not cheapness, and after 2000 the tidy relationship broke down and internationally it never held. The verdict a careful trader should carry: the Fed model is a *serviceable description of how the market has sometimes priced the stock-bond choice*, and a *false theory of fair value* — use it, if at all, as a sentiment gauge of relative positioning, never as a valuation anchor.

5.4 Rates → multiple compression, made quantitative

The most important fundamentals mechanism for a trader to be able to compute is how a change in rates moves the *multiple*, because that is the channel through which a macro surprise becomes an index move in a single afternoon. Use the justified multiple $P/E = b / (r - g)$ and put in illustrative but realistic numbers, anchored to the real 10-year yield from the dashboard in Section 6 (4.68% as of 2026-08). Take a payout ratio $b = 0.85$ (dividends plus buybacks), a long-run nominal cash-flow growth $g = 4.5\%$ (about trend nominal GDP), and a required equity return $r = 8.5\%$ (a nominal risk-free near 4.5% plus an ERP of about 4.0%). Then:

$$\text{Fair } P/E = 0.85 / (0.085 - 0.045) = 0.85 / 0.040 = \mathbf{21.3\times}.$$

Now raise the discount rate by **100 bp** — say the real risk-free rises a point, dragging r to 9.5% with growth unchanged:

$$\text{Fair } P/E = 0.85 / (0.095 - 0.045) = 0.85 / 0.050 = \mathbf{17.0\times}, \text{ a change of } 17.0/21.3 - 1 = \mathbf{-20\%}.$$

A one-point rise in the discount rate compresses the fair multiple by a fifth — and note the leverage: the multiple's sensitivity is governed by the *gap* $r - g$ in the denominator, so the same 100 bp does more damage when that gap is already small. If instead $r - g$ starts wide ($r = 11\%$, $g = 4.5\%$, gap 6.5%): $P/E = 0.85/0.065 = 13.1\times$, and +100 bp takes it to $0.85/0.075 = 11.3\times$, only **−13%**. This is the quantitative form of a truth traders state loosely as "long-duration assets are more rate-sensitive." The *effective duration* of the equity index is approximately $1/(r - g)$; in the first example that is $1/0.04 = \mathbf{25 \text{ years}}$, so a 100 bp move implies roughly a **−25%** first-order

price change (the exact figure is -20% because the relationship is convex, not linear). The two lessons: (1) in a *low-rate* world (small $r - g$) equities are *long-duration* and violently rate-sensitive, which is why 2022's rate shock compressed multiples so hard; and (2) because the mechanism runs through the *discount rate*, a rate move driven by *higher growth* (which also lifts g) compresses far less than an equal rate move driven by *higher inflation risk premium* (which lifts r alone) — the same 10-year yield can be benign or brutal for equities depending on *why* it rose.

6 Building a macro dashboard

Everything above is only useful if you can see it on one page and update it in minutes. This section ships a reproducible one-page macro dashboard built entirely from free data — the object a disciplined trader looks at before forming any macro view — and reads the current state off it.

6.1 The panels and the recipe

Figure 5 is eight panels, all FRED and Yahoo, all cached, with NBER recessions (USREC) shaded throughout so every series is read against the cycle: (1) *growth* — industrial-production YoY; (2) *inflation* — headline and core CPI YoY against the 2% line; (3) *labor* — the unemployment rate; (4) *policy* — the fed funds rate and the 10-year yield; (5) *cycle clock* — the 10-year-minus-3-month term spread, whose sign is the classic recession lead; (6) *financial conditions* — the weekly NFCI against its zero average; (7) *equities* — the S&P 500 on a log scale with its 200-day average; and (8) *drawdown* — the S&P's decline from its running maximum. The window is 2000 to present.

One-page macro dashboard - all series free (FRED + Yahoo), recessions shaded

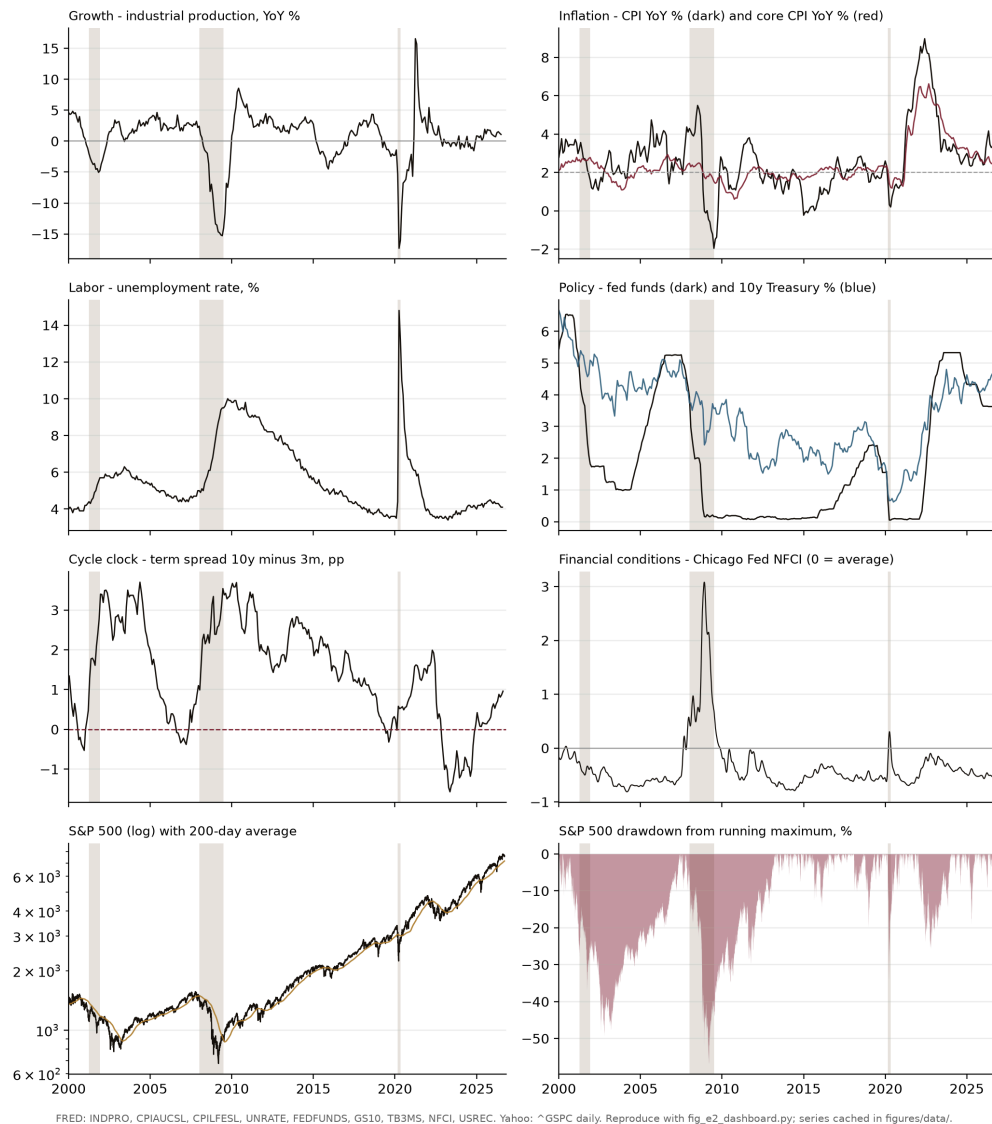


Figure 5. A one-page macro dashboard from free public data, 2000-2026. Eight panels — growth (industrial production YoY), inflation (headline and core CPI), labor (unemployment), policy (fed funds and 10-year), the term-spread cycle clock, the NFCI, the S&P 500 with its 200-day average, and the S&P drawdown — with NBER recessions shaded throughout. All series are FRED or Yahoo and cached for offline reproduction. Reproduce with `figures/fig_e2_dashboard.py`.

The recipe is nine FRED series (INDPRO, CPIAUCSL, CPILFESL, UNRATE, FEDFUNDS, GS10, TB3MS, NFCI, USREC) plus the Yahoo ^GSPC daily series, each pulled from its public endpoint and cached under `figures/data/`. Nothing here is proprietary, and the whole page regenerates from one script (`fig_e2_dashboard.py`).

6.2 Reading the current state

The latest values, printed by the script: **industrial-production YoY +1.08% (2026-07)** — growth positive but soft; **CPI YoY +3.35%, core CPI +2.45% (2026-08)** — inflation still above the 2% target, with core the stickier of the two; **unemployment 4.1% (2026-08)** — low by historical standards; **fed funds 3.63%, 10-year 4.68% (2026-08)**; **term spread +0.96 pp (2026-08)** — the 10y-minus-3m curve has *re-steepened* to clearly positive after its long inversion; **NFCI -0.560 (2026-09-11)** — financial conditions comfortably *looser* than average; and the **S&P 500 at 7,638 (2026-09-17), +6.4% above its 200-day average, only 2.07% below its record**. Synthesized into one line — the kind of read the dashboard exists to produce: *late-cycle but not yet contracting; growth soft-positive, inflation sticky above target, labor still firm, policy modestly restrictive with a re-steepened curve, financial conditions loose, and equities near highs*. That is precisely the configuration the Section 2 classifier labels "Deflation" (both growth- and inflation-momentum decelerating) while the *levels* remain expansionary — a reminder that a momentum-based regime label and a level-based dashboard read answer different questions, and a trader needs both. Three recessions are shaded since 2000 (2001, 2008–09, 2020), which is worth keeping in view for the next section's point about sample size: three cycles is *not* a large training set.

7 Macro → volatility and flows

Macro sets the slow-moving backdrop; the volatility factors of the volatility-surface paper and the flow calendar of the flow-landscape paper are the fast-moving surface where that backdrop becomes tradeable. This section maps the regime and reaction-function objects onto those faster layers. It is, by design, the most cross-referential section and the one that leans hardest on mechanism and practitioner consensus rather than fresh own-computation — a limitation flagged here and again in Section 8.

7.1 Regime maps onto the volatility factors

The volatility complex is not a single number; it is a set of factors — the level (VIX), the term structure (VIX versus VIX3M), the vol-of-vol (VVIX), skew, and cross-asset correlation — that the volatility-surface paper treats in detail. Macro regime shifts the *base rate* of each. In a Goldilocks state (growth up, inflation down) realized volatility tends to be low and the VIX term structure steeply upward-sloping (contango), the friendly configuration for short-vol and carry. Stagflation and the disinflationary-slowdown quadrants are where the term structure is most prone to *inversion* (backwardation) and where correlation spikes — the states in which diversification fails and the 60/40 portfolio (Section 2.4's negative TLT-in-Stagflation cell) breaks. The reaction-function gap of Section 3 is a *vol-of-vol* input: a large rule-minus-policy gap

(as now, +2.68 pp) means the policy path is contested, which is exactly when a data surprise can force a large repricing — the macro source of VVIX. The honest caveat, consistent with Section 2.4, is that these are *distributional* statements — regime raises or lowers the *odds* of a vol state, inside wide bands — not switches, and the historical counterexamples (calm stagflation stretches, violent goldilocks air-pockets) are numerous.

7.2 Regime and the flow calendar

The flow calendar of the flow-landscape paper — monthly and quarterly options expiration, the charm and vanna flows into expiry, the 0DTE cycle, and index rebalances — is a *mechanical* overlay that runs regardless of macro. Macro interacts with it in one specific way: it sets the *sign and size of the gamma backdrop* that those flows act through. In a low-vol, positive-gamma regime (often Goldilocks), dealer hedging is mean-reverting and the flow calendar produces the familiar pinning and drift-to-max-pain; in a high-vol, negative-gamma regime (often the stress quadrants), the same calendar dates become accelerants rather than dampeners. The macro layer does not create the flows; it decides whether they stabilize or destabilize. This is the cleanest statement of how the Context pillar meets the Microstructure pillar: *macro is the field, flows are the forces, and the field determines the sign of the force.*

7.3 Event-volatility around scheduled releases

The most direct macro-to-vol link is the scheduled release — CPI, non-farm payrolls, and the FOMC decision. Two robust, well-documented facts organize it (Lucca & Moench 2015 on the FOMC drift; general event-study evidence). First, implied volatility *builds into* a known event and *collapses after* it — the "vol crush" — because the event resolves a known chunk of uncertainty on a known date; options that straddle a CPI print carry a visible event premium that decays to zero the moment the number lands. Second, the *size* of the post-release move scales with the *surprise* relative to consensus, not with the level of the number — a hot CPI that was fully expected moves little, a mild one that was not expected can move a lot. The reaction-function gap sets the *asymmetry* of that surprise (Section 3.4): with the rule far above policy, the market is positioned for the data to lean hawkish, so a soft print carries more repricing power than a hot one. The practical discipline is to know the release calendar cold, to treat the days around CPI/NFP/FOMC as *elevated-event-vol* days whose realized move will most often *undershoot* the implied straddle (the short-event-vol edge) except when the surprise is large (the fat tail that makes the edge dangerous) — the same "sell the crush, respect the tail" structure the NFCI extreme showed in Section 4. Because this section ships no figure of its own, its quantitative claims are inherited from the volatility-surface paper and the cited literature and should be read as *plausible and mechanism-consistent, not independently validated here.*

8 Limits and humility

This section is not a disclaimer bolted on at the end; it is the load-bearing part of a Context-pillar paper. Macro is where confident backtests go to die, and the reasons are specific and knowable.

8.1 Regime labels are lagging and revised

Every label in Section 2 was computed from *final-vintage* data — today's fully revised INDPRO and CPIAUCSL. But the values a trader saw *in real time* were the *first prints*, which are revised, sometimes heavily. Industrial production and the national accounts undergo monthly revisions and periodic benchmark rewrites that can move a data point by more than the three-month momentum threshold that decides a label. So a month we now confidently call "Reflation" may have looked like "Stagflation" to someone standing in it with the data then available — and the reverse. The Section 2 classifier is therefore honest about *what each month was* and dishonest about *what a trader could have known*, and the lagged ($t-2$) table in Section 2.5 only fixes the *publication lag*, not the *revisions*.

8.2 The vintage trap: ALFRED versus FRED

This is the single most important technical point in the paper, and it has a name and a tool. FRED serves the *latest revised* value for every historical date; ALFRED (the "Archival" FRED) serves the value *as it was actually published at each point in time* — the *vintage*. A macro backtest built on FRED is silently using data that did not exist when its trades were supposedly placed: the 2008-Q4 GDP figure a strategy "traded on" in January 2009 is not the number FRED shows today. The magnitudes are not trivial: 2008-Q4 real GDP was first reported (the advance estimate) at roughly -3.8% annualized and was later revised down to around -8.5% after benchmark revisions — a gap of nearly five percentage points between what a real-time trader saw and what a FRED-based backtest "sees" now. A strategy that "went defensive when growth turned sharply negative" looks prescient on revised data and would have fired far later and weaker in real time. Multiply that across every macro series a strategy touches and the real-time edge can vanish entirely. This is *look-ahead bias* of the most insidious kind, because it hides inside data that feels legitimately historical. Orphanides showed this is not academic: reaction-function estimates computed on *real-time* (vintage) data differ *materially* from the same estimates on revised data, precisely because the real-time output gap the Fed actually saw was often very different from the one we now reconstruct (Orphanides 1998, 2001). The Taylor-rule gap in Section 3 uses revised data and therefore *overstates* how obviously "wrong" past policy was — the Fed was reacting to numbers that later got revised away. **The rule for any macro backtest is absolute: if the claim is about what was tradeable, the data must be vintage (ALFRED), not revised (FRED).** Every result in this paper that uses FRED levels is, by that rule, *descriptive of history as now known*, not a claim about real-time tradability, and each is labeled that way.

8.3 Overlapping windows and the tiny sample of cycles

Two sample problems corrupt macro significance. The first is *overlap*: the NFCI test in Section 4 used 2,901 weekly 21-day windows, but adjacent windows share 20 of their 21 days, so the effective number of independent observations is roughly the span divided by the horizon — closer to a few hundred than three thousand. Naive confidence intervals computed as if the windows were independent are *too narrow*, which is exactly why the non-overlapping subsample in Section 4.2 told a different story. The second, deeper problem is the *number of independent regimes*. We have 936 monthly labels, but they come from a handful of genuinely distinct macro epochs — the 1970s Great Inflation, the Volcker disinflation, the Great Moderation, the 2008 crisis and its zero-bound aftermath, the pandemic shock. The *effective* sample for any statement about "how assets behave in regime X" is the number of independent *episodes* of regime X, which is single digits, not hundreds. Three recessions since 2000 (Section 6.2) is the honest denominator for a lot of confident "late-cycle" trading. This is why every cross-quadrant difference in Section 2.4 came with a wide interval and a non-stationarity flag: the intervals are wide *because the true independent sample is small*, and pretending otherwise is the core macro-backtest lie.

8.4 Why macro-timing backtests lie

Collect the threads. A macro-timing backtest tends to overstate its edge because: (1) it uses *revised* data unavailable in real time (Section 8.2); (2) it matches signals to returns *contemporaneously*, ignoring the publication lag, and the edge shrinks the moment the lag is respected (Section 2.5); (3) it computes significance on *overlapping* windows, understating the standard errors (Section 8.3); (4) it implicitly has only a *few independent episodes* of any regime, so a "significant" result is often one or two cycles in disguise (Section 8.3); and (5) the classifier or threshold was frequently *chosen after seeing which split looked best* — the data-snooping the Quant track formalizes. Each of these pushes the *same direction*: toward a backtest that looks tradeable and is not. The constructive conclusion is not despair but discipline: use macro to shape the *distribution* — which tails are fat, which diversifiers are likely to fail, which surprises are asymmetric — and demand vintage data, respected lags, non-overlapping windows, and an out-of-period check before you let any macro signal *time* anything. Context is great for regimes and poor for timing, and a paper that pretended otherwise would be committing the error it was written to prevent.

8.5 A real-time-honesty checklist

Before any macro signal is allowed to *time* a position, run it against five questions, each mapped to a failure above. **Vintage?** — is every input the value that existed at the decision date (ALFRED), not the revised value (FRED)? **Lag?** — does the signal use only data published on or

before the entry, with the real release lag, not the reference month? **Overlap?** — are the confidence intervals computed on non-overlapping windows, or at least corrected for the horizon? **Episodes?** — how many *independent* macro episodes actually underlie the result, and would the headline survive dropping the single largest one? **Selection?** — was the classifier, threshold, or window chosen *before* or *after* seeing which split looked best? A signal that cannot answer all five cleanly is not necessarily wrong, but it has not earned a timing role — at most it earns a place in shaping the distribution, which is where Section 8.4 says macro belongs. The checklist is deliberately boring; boredom is the price of not fooling yourself.

9 Common misconceptions, glossary, and references

9.1 Common misconceptions

- **"We're in regime X, so buy/sell Y."** Regimes shift the odds inside wide, overlapping bands (Section 2.4) and turn over on a two-month median (Section 2.3). The label is a distribution, not a signal.
- **"The Taylor rule tells you the Fed's next move."** It tells you how far policy sits from a neutral benchmark and in which direction the pressure runs; it explicitly *failed* as a move-predictor across the Volcker 1980s (Section 3.3), when the Fed chose to sit 2.28 pp from the rule for a decade.
- **"Tight financial conditions mean lower returns."** On independent monthly windows over 1971–2026 they did not; tight-conditions months were followed by slightly *higher* average returns, because the NFCI is coincident with the bottom (Section 4.2). Only the crisis extreme is robustly informative.
- **"Stocks are cheap when the earnings yield beats the bond yield" (the Fed model).** It compares a real quantity to a nominal one — a money illusion — and its historical "fit" was an inflation-driven artifact of 1965–2000 (Section 5.3).
- **"Higher rates are bad for stocks."** *Why* they rose decides everything: rates up on higher growth (g rises with r) compress multiples far less than rates up on inflation risk premium (r rises alone) (Section 5.4).
- **"My macro backtest works."** Check whether it used revised data, matched signals contemporaneously, computed overlapping-window significance, and rests on a handful of independent cycles. If any is true — and usually several are — the edge is probably an artifact (Section 8).
- **"The NFCI/Taylor gap is a level to trade."** They are *state descriptors*. Their value is in shaping the distribution and the asymmetry of surprises, not in a mechanical entry.

9.2 Glossary

- **Growth×inflation quadrant** — a four-state regime scheme (Reflation, Goldilocks, Stagflation, Deflation) defined by the *direction* of growth and inflation.
- **Momentum of the YoY rate** — the three-month change in a year-over-year rate; positive = the annual rate is accelerating. Distinguishes "rising" from "high."
- **Reaction function** — a rule mapping the state of the economy (inflation, output/employment gaps) to the policy rate.
- **Taylor rule** — $i = r^* + \pi + 0.5(\pi - \pi^*) + 0.5 y$; the canonical fixed-coefficient reaction function.
- **Taylor principle** — for stability, nominal rates must rise *more* than one-for-one with inflation, so the *real* rate rises when inflation does.
- **Neutral (natural) rate r^*** — the real policy rate that neither stimulates nor restrains an economy at potential.
- **Output gap / Okun's law** — the percent deviation of output from potential; Okun's law links it to the unemployment gap with a coefficient of about 2.
- **Dual mandate** — the Fed's twin objectives of price stability and maximum employment, encoded as the two response terms of the rule.
- **Data-dependent** — a reaction function whose coefficients the bank reserves the right to change with the state; not the absence of a rule.
- **Financial-conditions index (FCI/NFCI)** — a standardized aggregate of many risk/credit/leverage measures; zero = historical average, positive = tighter.
- **Equity risk premium (ERP)** — the extra expected return demanded for equities over the risk-free asset; *historical* (realized) versus *implied* (from a forward model).
- **Fed model** — the heuristic equating fair value with earnings yield $\approx$ 10-year yield; critiqued as a real-versus-nominal money illusion.
- **Justified multiple** — $P/E = b/(r - g)$; the fair price-earnings ratio implied by payout, discount rate, and growth.
- **Multiple compression** — a fall in the P/E driven by a rising discount rate; larger when $r - g$ is small (long equity duration).
- **Equity duration** — the sensitivity of the index to the discount rate, $\approx 1/(r - g)$.
- **Vintage / real-time data (ALFRED)** — the value of a series *as originally published* at a past date, before revisions; the correct data for a tradability claim.
- **Look-ahead bias** — using information in a backtest that was not available at the modeled decision time; the vintage trap is its most insidious form.

- **Non-stationarity** — the property that a series' statistical behavior changes over time, so past cell means need not describe the future.
- **Overlapping windows** — forward-return windows that share days; they understate standard errors and overstate significance.
- **Vol crush** — the collapse of implied volatility once a scheduled event resolves its uncertainty.

Evidence basis

Every substantive claim in this paper rests on one of three kinds of basis, distinguished here so a reader can tell at a glance which is which — **own reproducible computation** (recomputed from free data or a self-contained model, reproducible via the cited script), **literature** (peer-reviewed article or textbook, cited inline), and **practitioner consensus** (desk practice with no clean citation, treated as the weakest). Nothing here is called *validated* on external sources alone; the stronger word requires an own out-of-sample computation.

Claim / result	Basis
Growth×inflation regimes split near-evenly (24–27% each) over 1948–2026 and turn over on a ~2-month median (61.6% same-label persistence) (Figure 1)	Own reproducible computation — figures/fig_e2_quadrant.py
Mean monthly asset returns by quadrant have wide, overlapping 95% CIs, and respecting the publication lag shrinks the signal (TLT signs flip in three of four cells) (Figure 2)	Own reproducible computation — figures/fig_e2_quadrant_assets.py
A fixed-coefficient Taylor rule tracks the funds rate over 1960–2026 (correlation 0.749, mean absolute gap 1.95 pp); the current rule-minus-policy gap is +2.68 pp and the largest was +9.93 pp in March 2022 (Figure 3)	Own reproducible computation — figures/fig_e2_taylor.py
The NFCI is a weak monthly return timer — level-versus-forward-return correlation -0.055 , and the loose/tight ordering reverses on non-overlapping windows (Figure 4)	Own reproducible computation — figures/fig_e2_nfci.py
The one-page dashboard's current read — late-cycle, growth soft-positive, inflation sticky above target, curve re-steepened, conditions loose, equities near highs (Figure 5)	Own reproducible computation — figures/fig_e2_dashboard.py
The Taylor rule ($i = r^* + \pi + 0.5(\pi - \pi^*) + 0.5y$) and the Taylor principle	Literature — (Taylor 1993)
Real-time (vintage) reaction-function estimates differ materially from the same estimates on revised data	Literature — (Orphanides 1998, 2001)
The NFCI is a standardized aggregate of 105 risk/credit/leverage measures (mean zero, unit sd) with a three-subindex decomposition	Literature — (Brave & Butters, Chicago Fed)
The Fed model commits a real-versus-nominal money-illusion error and its 1965–2000 "fit" was an inflation-driven artifact	Literature — (Asness 2003)
Corporate margins mean-revert (so extrapolating peak margins is a valuation error) and the equity risk premium is a derived, assumption-sensitive quantity	Literature — (Damodaran; Ilmanen 2011)
The pre-FOMC announcement drift — equities drift up ahead of scheduled FOMC decisions	Literature — (Lucca & Moench 2015)

Claim / result	Basis
Regime sets the base rate of volatility states and the sign of the gamma backdrop the flow calendar acts through, and implied vol builds into events then collapses (the "vol crush"); inherited mechanism claims, not validated here	Practitioner consensus — not independently verified

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